

Tech Team Intern

AI Chatbot

Vector Search

Prompt Engineering

Document Q&A;

OVERVIEW

Build a simple AI chatbot with a web UI that can answer questions about Verite Research publications. The chatbot should have a defined persona, maintain the conversation within a session, and retrieve answers from a vector database built from the documents you provide. You are not expected to build a production system — we want to see how you think, how you write prompts, and how you approach a problem from scratch.

Download any **three publications** from the Verite Research catalogue and use them as your knowledge base:

<https://www.veriteresearch.org/services-and-products/research-outputs/>

WHAT TO BUILD

Your application must do the following:

- 01 Load and extract text from your 3 chosen Verite PDFs
- 02 Chunk and embed the text, and store it in a vector database of your choice
- 03 Give the chatbot a name and a clear persona — define this in a system prompt
- 04 Accept user questions via a web UI, retrieve relevant chunks from the vector DB, and answer using that context
- 05 Maintain conversation history within a session (the bot remembers earlier messages)
- 06 Cite which document an answer came from
- 07 Politely decline questions unrelated to Verite's work
- 08 Respond naturally to greetings and small talk without triggering a vector search

Bonus (optional)

- + Hybrid search (vector + keyword)
- + Show source chunk and page number alongside the answer
- + Long-term memory across sessions

HOW THE CHATBOT SHOULD BEHAVE

The bot must handle each of these message types correctly:

Message type	What the bot does	Example
Greeting / small talk	Responds naturally, no vector search	"Hi!" — greets back, introduces itself
Question about Verite content	Searches vector DB, answers with citation	"What does Verite say about forced labour?"
Follow-up question	Uses conversation history for context	"Can you say more about that?"
Out-of-scope question	Politely declines and redirects	"Who is Virat Kohli?" — declines

In your PROMPTS.md, explain how you handle borderline questions — e.g. "What is forced labour?" It is a general term but directly relevant to Verite's work. There is no single right answer, but document the choice you made and why.

TOOLS & STACK

Use whatever you are comfortable with. These are common starting points, not requirements:

Area	Options
Vector DB	ChromaDB, FAISS, Qdrant, Pinecone, Weaviate, pgvector
LLM / API	OpenAI GPT-4o, Anthropic Claude, Google Gemini, Mistral
PDF extraction	pypdf, pdfplumber, PyMuPDF
Language / Framework	Python (recommended), Node.js, or any language you know well
UI	Streamlit, Gradio, or plain HTML/JS

SUBMISSION

Send us a link to your GitHub repository (public, or private with access shared). It should contain:

- 01 All source code and a README explaining how to run it locally
- 02 A PROMPTS.md explaining your system prompt and the decisions you made
- 03 The 3 Verite publications you used (title, year, URL) — do not commit the PDFs
- 04 3 to 5 sample questions and the answers your chatbot gave

Deployment is not required — a working local setup is enough. Do not commit API keys; use environment variables and include a **.env.example** file.

We care more about clear thinking and clean code than a fancy stack. Good luck.